RAG Knowledge Base — Sources

This document describes the source corpus behind the Sep Agent's opt-in retrieval-augmented generation (RAG) layer: what is in the production index, what is deliberately kept out, and why.

How the index is built (one-line recap)

Only the **top-level PDFs** in `rag_sources/` are ingested (the ingestion glob is non-recursive, so anything in a subfolder is automatically excluded without being deleted). Each PDF is chunked with a header-aware splitter, passed through a quality gate that drops boilerplate (bibliographies, footers, garbled text), and embedded with a local sentence-embedding model. The result is saved as the production index under `results/rag_knowledge_base/`.

Production index: 3 sources / 83 chunks (2 chunks dropped by the quality gate).

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1. Sources IN the production index (diagnostic index)

These are curated for **clean, retrieval-oriented clinical criteria** — the genre that actually helps the aggregator ground its synthesis.

Source file	Chunks	What it is	What it contributes
`cardiac_ai_internship.pdf`	59	The project's own consolidated clinical reference notes (single-column, clean text).	General diagnostic criteria and normal ranges across ECG / MRI / clinical findings; the backbone of the index.

`ESC_2021_Heart_Failure_RAG_Source.pdf`	13	A cleanly reformatted, single-column extract of ESC 2021 Heart-Failure guideline content.	Ejection-fraction categories (HFrEF ⩽40 %, HFmrEF 41–49 %, HFpEF ⩾50 %), NYHA functional class, advanced-HF criteria.
`AHA_ACCF_HRS_2009_ECG_Chamber_Hypertrophy_RAG_Source.pdf`	11	A cleanly reformatted extract of the AHA/ACCF/HRS 2009 recommendations on ECG chamber-hypertrophy interpretation.	Voltage-based chamber-hypertrophy (LVH) criteria that back the Sokolow–Lyon flag.

Why these two guideline files are re-authored extracts, not the raw PDFs.
A raw guideline PDF (complex two-column layout) extracts with spaces stripped (~26 % of "words" merged), which degrades retrieval and is unreadable to the model. The two guideline sources above were therefore reformatted as clean, single-column text (<1 % merged words) specifically for reliable retrieval.

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2. Sources kept but EXCLUDED from the index (subfolders)

These are preserved for provenance / future use but are **out of the ingestion glob** on purpose. The controlled A/B study (`results/rag_ab_test/rag_utility_ab_synthesis.md`) showed why: dataset-description and epidemiology papers are dense lists of annotation codes and population statistics that dominate similarity retrieval and crowd out real clinical criteria, so they must not sit in the *diagnostic* index.

Subfolder	Files	Nature	Why excluded from the diagnostic index
`dataset_docs/`	PTB-XL (Wagner 2020), ACDC (Bernard 2018), Alqudah 2026 multimodal	Dataset-description papers	Cited for the datasets themselves; their annotation-code lists pollute diagnostic retrieval.

`epidemiology/`	PURE global CVD (Lear 2025), NHANES CVD (Ahiduzzaman 2025), UK Biobank ethnicity (Razieh 2022), social determinants (Kreatsoulas 2010)	Population CVD epidemiology	Not diagnostic criteria; reserved for a possible future, separately scoped risk-context index.
`guidelines_raw/`	Raw `2021_ESC_Guidelines_for_HF.pdf`	Full raw guideline PDF	Two-column layout extracts garbled (~26 % merged words); replaced by the clean ESC extract above.
`New/`	Recent cardiology papers (aficamten trials, aortic stenosis, HF device therapy, etc.)	Recent literature, staged	Newly added, not yet curated/ingested; left in a subfolder so it is out of the index until reviewed.

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3. Provenance and usage note

The guideline extracts (`ESC_2021_*` , `AHA_ACCF_HRS_2009_*`) are short, reformatted excerpts of published clinical-guideline content, prepared **for local research retrieval only**. Guideline text is copyrighted by its issuing societies; these extracts are used on-device, are not redistributed, and are not a substitute for the official guideline documents. The dataset and epidemiology papers retain their original authorship as cited in the report bibliography.

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4. Design conclusion (from the A/B study)

Curation of knowledge-base **content** — not the retrieval mechanism or the embedding model — is the dominant factor in whether RAG helps: with the clean, targeted diagnostic index above, the aggregator cites real criteria (e.g. the HFrEF ⩽40 % threshold); with an unfiltered index dominated by dataset/epidemiology text, it cites boilerplate or nothing. Full evidence: `results/rag_ab_test/rag_utility_ab_synthesis.md` . Adding one or two more clean, authoritative guideline extracts to `rag_sources/` (top level) would strengthen the index; a rebuild picks them up automatically.